

SLIP 20

Q.1) Write a C program that illustrates suspending and resuming processes using signals

[10 Marks]

```
#include<stdio.h>
#include<unistd.h>
#include<signal.h>
#include<sys/wait.h>

int main()
{
    pid_t pid;
    pid=fork();
    if(pid < 0)
    {
        perror("fork");
        exit(455);
    }

    if(pid==0)
    {
        while(1)
        {
            printf("child process is running : \n");
            sleep(3);
        }
    }
    else
    {
        printf("\nparent : child process id is : %d\n",pid);
        sleep(5);

        printf("parent : suspending parent process\n");
        kill(pid,SIGSTOP);
        sleep(5);

        printf("parent :child process is resuming \n");
        kill(pid,SIGCONT);
        sleep(5);

        printf("parent :child process is terminating\n");
        kill(pid,SIGTERM);
        wait(NULL);
    }
}
```

```

        printf("parent : child process is terminated \n");
    }
}

```

Q.2) Write a C program to Identify the type (Directory, character device, Block device, Regular file,

FIFO or pipe, symbolic link or socket) of given file using stat() system call.

[20 Marks]

```

#include <stdio.h>
#include <stdlib.h>
#include <sys/stat.h>
int main(int argc, char *argv[]) {
    if (argc != 2) {
        printf("Usage: %s <filename>\n", argv[0]);
        return 1;
    }
    struct stat st;
    if (stat(argv[1], &st) == -1) {
        perror("stat");
        return 1;
    }
    printf("File: %s\n", argv[1]);
    printf("File type: ");
    if (S_ISREG(st.st_mode)) printf("Regular file\n");
    else if (S_ISDIR(st.st_mode)) printf("Directory\n");
    else if (S_ISCHR(st.st_mode)) printf("Character device\n");
    else if (S_ISBLK(st.st_mode)) printf("Block device\n");
    else if (S_ISFIFO(st.st_mode)) printf("FIFO/pipe\n");
    else if (S_ISLNK(st.st_mode)) printf("Symbolic link\n");
    else if (S_ISSOCK(st.st_mode)) printf("Socket\n");
    else printf("Unknown\n");
    return 0;
}

```